

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

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COURSE 04 OF 12
CURRICULUM GUIDE

Data Science for Teens

Turning real data into real insight

GRADE BAND

Grade 9–12

DIFFICULTY

Intermediate

DURATION

14 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

Data science is how modern decisions get made. Students learn to collect, clean, visualize and analyze real datasets using Python's professional data toolkit — and to tell a clear story with what they find.

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Prerequisites: Python for Kids & Teens.

LEARNING OUTCOMES

What Students Will Learn

- Load, clean and explore real-world datasets with pandas
- Create meaningful visualizations with matplotlib and seaborn
- Apply descriptive statistics to summarize data
- Ask and answer real questions using a data-analysis workflow
- Understand how clean data feeds machine learning
- Communicate findings clearly in a written data report

WHAT WE'LL USE

Tools & Technologies

Python pandas NumPy matplotlib / seaborn Jupyter Notebook

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · Intro to Data Science & Jupyter	What data scientists actually do; the notebook workflow.
02 · Working with pandas	DataFrames, reading CSVs, filtering, sorting.
03 · Data Cleaning	Handling missing values, duplicates and data types.

MODULE	FOCUS
04 · Data Visualization I	Bar, line and scatter charts with matplotlib.
05 · Data Visualization II	Seaborn, distributions and correlations.
06 · Descriptive Statistics	Mean, median, mode, spread and summary statistics.
07 · Exploratory Data Analysis Project	Asking and answering original questions from a dataset.
08 · From Data to Decisions & Capstone	Data storytelling; final analysis report.

LEARNING BY BUILDING

Projects Students Will Build

- Real Dataset Exploration — Sports, movies or climate data — student's choice.
- Data Visualization Notebook — A polished, annotated set of visual insights.
- Capstone Data Report — A written and visual analysis of a self-chosen dataset.

BEYOND THE SYNTAX

Skills Developed

Data wrangling Visualization Statistical thinking Analytical communication

PORTFOLIO OUTCOME

A polished data-analysis notebook and report, ready to share on GitHub.

RECOMMENDED NEXT STEP

Agentic AI for Teens, or Generative AI for Teens